
Kernel-Preconditioned Burer–Monteiro Flow: A Spectral Implicit Bias for Wide Networks Trained with Triplet Loss

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Abstract

Neural metric learning trains an embedding network with a triplet or contrastive loss, but which similarity geometries gradient descent can actually fit on a finite training horizon is poorly understood. We address this by bridging two theoretical lenses — the neural tangent kernel (NTK) for wide-network gradient dynamics, and the Burer–Monteiro factorisation for low-rank semidefinite optimization — in a way that has not been combined for primal feature-space training: gradient flow on a wide embedding network induces a closed-form *kernel-preconditioned matrix flow* on the Gram matrix of the embeddings, with the NTK at initialization as the preconditioner. The resulting dynamics exhibit a *quadratic* spectral implicit bias — per-mode displacement of the Gram matrix scales as the square of the corresponding NTK eigenvalue, distinct from the linear-in-eigenvalue rates known for scalar NTK regression — which pins down the effective hypothesis class on bounded training horizons: target structures aligned with the kernel’s top eigenspace are reachable, while those concentrated in its bottom eigenspace remain near initialization. We verify the slope ≤ 2 bound across synthetic data, Fashion-MNIST, frozen-feature CIFAR-10, and an end-to-end CNN trained from scratch. The same analysis prescribes a one-line algorithmic correction — applying the inverse kernel to the embedding-side gradient — which improves hard-negative retrieval but not easy retrieval, exactly the regime where the theory predicts an effect.

1 Introduction

Neural metric learning trains an embedding $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^k$ so that $\|f_\theta(x) - f_\theta(y)\|^2$ encodes a desired similarity geometry. The standard recipe is to minimize a triplet or contrastive loss — a twice-differentiable function of the embeddings’ pairwise inner products — so that the loss depends on θ only through the Gram matrix $G = FF^\top$ of the embeddings. This Gram-matrix view is what makes metric learning amenable to two distinct theoretical frameworks that otherwise bear on it from very different angles. The *neural tangent kernel* [Jacot et al., 2018, Lee et al., 2019] describes sufficiently wide networks as kernel methods governed by a fixed kernel Θ , and yields linear ODEs for the network’s predictions under gradient flow. The *Burer–Monteiro factorization* [Burer and Monteiro, 2003, 2005], central to low-rank semidefinite programming, parameterizes a positive semidefinite matrix X as $X = UU^\top$ and uses U as the optimization variable; the resulting gradient flow on X takes the form $\dot{X} = -2(EX + XE)$.

The two frameworks are usually treated in isolation. We connect them by observing that an embedding network with twice-differentiable loss on $G = FF^\top$ is itself a Burer–Monteiro factorization,

but in the non-linear feature space induced by the network rather than the linear parameterization $G = UU^\top$ of classical low-rank SDP. The NTK enters this picture as an explicit *kernel preconditioner* on the BM dynamics. Concretely, in the NTK regime the network’s Gram matrix evolves according to

$$\dot{G} = -2(KEG + GEK), \quad E = \partial L / \partial G, \quad (1)$$

which reduces to the classical BM flow when $K = I$. We call (1) the *kernel-preconditioned Burer–Monteiro* (kBM) flow.

The equivalence is itself a conceptual bridge, but the load-bearing use of it is to extract a *spectral implicit bias* that does not follow from the NTK ODE alone: in K ’s eigenbasis, mode- (i, j) evolution is multiplied by $\lambda_i + \lambda_j$, and at NTK initialization per-mode displacement scales *quadratically* in λ_i . The trajectory of $G(t)$ is therefore confined to K ’s top eigenspaces on bounded training horizons, regardless of where the loss would otherwise push it. We verify the slope- ≤ 2 bound across synthetic and real-image setups, and derive a one-line K^{-1} correction with a precise regime-of-effect prediction that the data confirm.

Contributions. (i) *kBM equivalence* (Theorems 2–4): single-layer and depth- L ReLU gradient flow on $L(G)$ obeys $\dot{G} = -2(KEG + GEK)$, recovering classical BM when $K = I$; (ii) *spectral implicit-bias theorems*: a model-free linear-in- λ rate ordering (Theorem 6) sharpens to a quadratic prediction in the NTK regime (Theorem 8), yielding an effective hypothesis class of rank $r(T, \delta)$ (Corollary 10); (iii) *multi-seed numerical verification* on synthetic data, Fashion-MNIST, frozen-feature CIFAR-10, and an end-to-end CNN, with slopes spanning 1.15–2.19 (all ≤ 2); (iv) *theory-derived correction*: a one-line K^{-1} preconditioner flattens the slope across all settings and improves hard-negative CIFAR-10 Recall@10 by +3.2 pp ($p = 0.004$) with no gain on easy retrieval — exactly the regime of effect theory predicts; (v) the bias is empirically *distinct from Razin–Cohen rank bias*. All results assume the standard isotropic NTK approximation; we do not claim a quantitative generalisation bound (Section 5).

2 Related Work

NTK theory. The NTK framework [Jacot et al., 2018] reduces wide-network training to kernel regression; Lee et al. [2019] extend the convergence to any depth L and give the standard finite-width concentration rate. Chizat et al. [2019] formalises the lazy/rich dichotomy.

Burer–Monteiro factorization, and the closest BM-meets-NN prior art. Burer and Monteiro [2003, 2005] introduced $X = UU^\top$ for SDPs; Boumal et al. [2016], Bhojanapalli et al. [2016], Ge et al. [2016], Park et al. [2017] characterise the resulting non-convex landscape. Sahiner et al. [2022] apply BM to the *dual* variable of the Pilanci–Ergen convex reformulation of two-layer networks for computational scalability. Our connection is distinct: we identify that *primal* gradient flow on a wide feature network induces a kernel-preconditioned BM flow on the embeddings’ Gram matrix $G = FF^\top$, with the NTK as the explicit preconditioner. To our knowledge no prior work derives this primal-Gram kBM equivalence for metric learning or extracts a spectral implicit bias from it.

Implicit bias. Gunasekar et al. [2017], Arora et al. [2019], Razin and Cohen [2020] characterise an implicit *rank* bias in matrix and tensor factorisation. Soudry et al. [2018], Lyu and Li [2020] characterise asymptotic margin direction. Our bias is in the spectrum of the NTK, not the rank of G nor the parameter direction; we demonstrate the dissociation in Section 4.7.

NTK eigenvalue dynamics and spectral bias. Bordelon et al. [2020], Canatar et al. [2021] characterise mode-wise generalisation error in NTK *kernel regression* on scalar targets, showing rates linear in the kernel eigenvalue. Our object is the *matrix-valued* Gram matrix G under triplet loss, and our dynamics yield a *quadratic* per-mode displacement upper-bound: the extra factor of λ comes from the Funk–Hecke alignment between G_0 and K at NTK initialization (Proposition 7), which has no analog in scalar regression. Rahaman et al. [2019], Basri et al. [2020] study a different “spectral bias” — input-Laplacian frequencies in scalar regression — distinct from our NTK-eigenbasis bias on a Gram matrix.

Kernel-spectrum-modifying preconditioners. The closest algorithmic neighbour is Amir et al. [2023], who modify the NTK spectrum directly to control inductive bias in *scalar* (regression / classification) wide-network training. Their goal is to use the modified spectrum to enforce a desired function-space prior; ours is to cancel an existing primal-Gram-matrix spectral bias derived from the kBM equivalence. The two interventions both apply a kernel-dependent preconditioner to the gradient, but operate on different objects (their scalar predictor vs. our Gram matrix), under different theoretical analyses (their function-space spectral-bias modification vs. our matrix-valued kBM flow), and target different regimes of effect (their feature-imposition for arbitrary priors vs. our hard-negative bottom-eigenspace recovery for metric learning). To our knowledge no prior work derives the primal-Gram kBM equivalence or characterises the slope- ≤ 2 bound for matrix-valued NTK dynamics.

Metric-learning empirics. Schroff et al. [2015], Weinberger and Saul [2009], Movshovitz-Attias et al. [2017], Wang et al. [2017] develop triplet metric learning empirically; our convex-reformulation neighbour is Pilanci and Ergen [2020].

3 Theoretical Framework

3.1 Setup

Let $X \in \mathbb{R}^{N \times d}$ be a data matrix with rows x_i . We denote by $F = f_\theta(X) \in \mathbb{R}^{N \times M}$ the embeddings produced by a depth- L fully-connected ReLU network of hidden width M with parameters θ ; we use the last hidden-layer activations as the embedding so the embedding dimension equals the hidden width. The corresponding Gram matrix is $G = FF^\top \in \mathbb{R}^{N \times N}$. We consider any twice-differentiable loss $L : \mathbb{R}^{N \times N} \rightarrow \mathbb{R}$ on G ; triplet loss is the running example. Write $E := \partial L / \partial G$.

We use the standard NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ at each layer and no output normalization; under this convention G_{ij} is $\Theta(1)$ at initialization and our expressions are scale-free.

3.2 Single-layer kBM equivalence

Definition 1 (NTK at the data). *The Jacobian of the embedding is $J(x; \theta) = \nabla_\theta f_\theta(x)$. The (output-summed) NTK at x_i, x_j is $\kappa(x_i, x_j) := \text{tr}(J(x_i)J(x_j)^\top)/M$. We write $K_{ij} := \kappa(x_i, x_j)$ for the empirical kernel matrix.*

Theorem 2 (kBM equivalence, single layer). *For a single-layer ReLU network $F_i = \text{ReLU}(x_i^\top W)$ with $W \in \mathbb{R}^{d \times M}$ trained by gradient flow on $L(G)$, the Gram matrix G evolves, in the limit $M \rightarrow \infty$, as*

$$\dot{G} = -2(KEG + GEK).$$

For finite width M the same equation holds with the empirical kernel $K^{(M)}$ in place of K ; the gap $\|K^{(M)} - K\|_{\text{op}} = O_p(M^{-1/2})$ under standard assumptions [Jacot et al., 2018, Lee et al., 2019].

Proof sketch. Under gradient flow the Gram-Gram NTK $\Theta_{(ij),(kl)}^G := \langle \nabla_\theta G_{ij}, \nabla_\theta G_{kl} \rangle$ admits a four-term contraction $\kappa_{ik}G_{jl} + \kappa_{il}G_{jk} + \kappa_{jk}G_{il} + \kappa_{jl}G_{ik}$ in the $M \rightarrow \infty$ limit. Substituting into $\dot{G}_{ij} = -\sum_{kl} E_{kl} \Theta_{(ij),(kl)}^G$ and using $E = E^\top$ collapses the four terms into $KEG + GEK$. Full derivation: Appendix A. $\square$

Remark 3. *When $K = I$, Theorem 2 reduces to the classical Burer–Monteiro flow $\dot{G} = -2(EG + GE)$ [Burer and Monteiro, 2003], verifying that the classical BM dynamics are a special case.*

3.3 Depth- L extension

The depth- L NTK admits the recursive form of Jacot et al. [2018]: $\Theta^{(\ell)} = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \odot \Theta^{(\ell-1)}$ with $\Theta^{(1)} = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \odot \langle x, y \rangle$, where $\Sigma^{(\ell)}$ is the layer- ℓ activation kernel (the Cho–Saul arc-cosine kernel) and $\dot{\Sigma}^{(\ell)}$ its derivative [Cho and Saul, 2009].

126 **Theorem 4** (kBM equivalence, depth L). *For a depth- L fully-connected ReLU network in NTK*
 127 *parameterization, the Gram matrix G evolves under gradient flow as $\dot{G} = -2(K_L EG + GEK_L)$*
 128 *where K_L is the depth- L NTK from the Jacot recursion.*

129 *Proof sketch.* The four-term Gram-Gram NTK expansion from Theorem 2 follows from $G = FF^\top$
 130 alone and is depth-independent; replace κ by κ_L . Full derivation: Appendix A. $\square$

131 3.4 Spectral implicit bias

132 The kBM flow takes a particularly clean form in K 's eigenbasis, and this form yields the paper's
 133 central new result.

134 **Lemma 5** (kBM in K 's eigenbasis). *Let $K = U\Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_1 \geq \dots \geq$
 135 $\lambda_N \geq 0$. Define $\tilde{G} := U^\top GU$ and $\tilde{E} := U^\top EU$, and assume E is symmetric (true whenever L is a
 136 function of pairwise inner products of F , including triplet, contrastive, and pairwise losses). Then
 137 the kBM flow becomes*

$$\dot{\tilde{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda),$$

138 *and the (i, j) entry satisfies $\dot{\tilde{G}}_{ij} = -2(\lambda_i(\tilde{E}\tilde{G})_{ij} + \lambda_j(\tilde{G}\tilde{E})_{ij})$. By symmetry of $\tilde{E}$ and $\tilde{G}$, the*
 139 *diagonal mode satisfies $\dot{\tilde{G}}_{ii} = -4\lambda_i(\tilde{E}\tilde{G})_{ii}$, so the i th eigendirection of G relaxes at instantaneous*
 140 *rate proportional to λ_i . In particular modes with $\lambda_i = 0$ are stationary, and modes with small λ_i*
 141 *are correspondingly slow.*

142 *Proof.* Direct substitution of $K = U\Lambda U^\top$ and conjugation by $U^\top, U$. $\square$

143 **Theorem 6** (Spectral implicit bias, model-free linear bound). *Let $G(t)$ solve the kBM ODE $\dot{G} =$
 144 $-2(KEG + GEK)$ with $L \in C^2$. Suppose $\lambda_i = 0$ for some index i . Then $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$ for
 145 all t and all j . More generally, on any interval $[0, T]$ on which the trajectory satisfies the stability
 146 condition $\|\tilde{E}(s)\tilde{G}(s)\|_{op}, \|\tilde{G}(s)\tilde{E}(s)\|_{op} \leq C$ for all $s \in [0, T]$, every entry (i, j) obeys*

$$|\tilde{G}_{ij}(T) - \tilde{G}_{ij}(0)| \leq 2T(\lambda_i + \lambda_j)C.$$

147 *Proof sketch.* Frozen-mode case: rows/columns of Λ at index i vanish in Lemma 5's expression.
 148 Bound: integrate $|\dot{\tilde{G}}_{ij}| \leq 2(\lambda_i + \lambda_j)C$. Full proof: Appendix A. $\square$

149 3.5 From a linear bound to a quadratic prediction

150 The bound in Theorem 6 is linear in λ_i . In our experiments (Section 4.3) the empirical scaling of
 151 diagonal displacements $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ is consistently *quadratic* in λ_i across depths and across
 152 both synthetic and Fashion-MNIST data. The gap between linear bound and quadratic observation
 153 is closed by exploiting a structural property of NTK initialization: G_0 and K share approximate
 154 eigenstructure, so $\tilde{G}_0$ is approximately diagonal in K 's eigenbasis with diagonal entries proportional
 155 to the kernel spectrum.

156 **Proposition 7** (Approximate eigen-alignment at NTK init via Funk–Hecke). *For a single-layer*
 157 *ReLU network in NTK parameterization with width M and inputs drawn iid from a rotationally-*
 158 *invariant distribution on the unit sphere, $G_0 = F_0 F_0^\top$ converges as $M \rightarrow \infty$ to the layer-1 arc-*
 159 *coshine kernel $\Sigma^{(1)}$ [Cho and Saul, 2009]. Both $\Sigma^{(1)}$ and the NTK $K = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \odot \langle x_i, x_j \rangle$ are*
 160 *functions of the input correlation matrix $\langle x_i, x_j \rangle$ alone; under rotational invariance they share an*
 161 *asymptotic eigenbasis (the spherical harmonics on S^{d-1}). Assume the empirical kernel matrix $K^{(N)}$*
 162 *has spectral gap bounded below by some $g > 0$ across its N eigenvalues. Then $\tilde{G}_0 := U^\top G_0 U$ is*
 163 *approximately diagonal, and on the empirical-rank-supported finite range of harmonic frequencies*
 164 $l \in \{0, \dots, L_{\max}(N)\}$ *the diagonal entries satisfy $\tilde{G}_0[i, i] = c_i \lambda_i$ with $c_i \in [c_{\min}, c_{\max}]$ for finite*
 165 *constants $c_{\min} > 0$ and $c_{\max} \leq 1$.*

166 *Proof sketch.* Both $\Sigma^{(1)}$ and K are scalar functions of $\langle x_i, x_j \rangle$, so Funk–Hecke [Atkinson and
 167 Han, 2012, Thm. 2.22] diagonalises them in the spherical-harmonic basis with eigenvalues given
 168 by Gegenbauer integrals; the ratio $c_l \in (0, 1)$ and is bounded away from zero by the decay rates
 169 of Bach [2017]. Empirical concentration is $O(N^{-1/2})$ [Tropp, 2015] via Davis–Kahan. Full proof:
 170 Appendix A.4. $\square$

171 **Theorem 8** (Refined spectral implicit bias, quadratic prediction). *Let $G(t)$ solve the kBM ODE*
 172 $\dot{G} = -2(KEG + GEK)$ *with $L \in C^2$. Assume:*

173 (A1) $\tilde{G}_0$ *has diagonal entries $\tilde{G}_0[i, i] = c_i \lambda_i$ with $c_i \in [c_{\min}, c_{\max}]$, and the total off-diagonal*
 174 *mass is small in Frobenius norm: $\|\tilde{G}_0 - \text{diag}(\tilde{G}_0)\|_F = O(N^{-1/2})$ (justified by Proposi-*
 175 *tion 7);*

176 (A2) $\tilde{E}(s)$ *has uniformly bounded operator norm $\|\tilde{E}(s)\|_{op} \leq E_{op}$ for $s \in [0, T]$, and*
 177 $|(\tilde{E}(s))_{ii}| \leq E_{op}$ *entrywise on the diagonal.*

178 *Then, to leading order in T ,*

$$\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0) = -4T \lambda_i (\tilde{E}_0 \tilde{G}_0)_{ii} + O(T^2),$$

179 *and combining (A1) and (A2) via Cauchy–Schwarz gives*

$$\boxed{|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)| \leq 4c_{\max} T E_{op} \lambda_i^2 + O(T \lambda_i N^{-1/2}) + O(T^2).}$$

180 *The leading term scales as λ_i^2 , matching the empirical power-law slope ≤ 2 observed in Section 4.3.*

181 *Proof sketch.* By Lemma 5, $\dot{\tilde{G}}_{ii} = -4\lambda_i (\tilde{E} \tilde{G})_{ii}$. Linearising around $t = 0$ gives $\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0) =$
 182 $-4T \lambda_i (\tilde{E}_0 \tilde{G}_0)_{ii} + O(T^2)$. Decompose $\tilde{G}_0 = D + R$ where $D = \text{diag}(\tilde{G}_0)$ and R is the off-
 183 diagonal residual. Then $(\tilde{E}_0 \tilde{G}_0)_{ii} = (\tilde{E}_0)_{ii} c_i \lambda_i + (\tilde{E}_0 R)_{ii}$, and Cauchy–Schwarz with operator-
 184 norm $\|\tilde{E}_0\|_{op} \leq E_{op}$ and $\|R\|_F = O(N^{-1/2})$ from (A1) gives $|(\tilde{E}_0 R)_{ii}| \leq \|\tilde{E}_0\|_{op} \|R\|_F =$
 185 $O(N^{-1/2})$. Plugging these in yields the boxed bound. Full proof: Appendix A. $\square$

186 **Remark 9.** *Theorem 6 (linear bound) and Theorem 8 (quadratic bound) characterize the same*
 187 *phenomenon at different levels of refinement. The linear bound is model-free: it holds for any kBM*
 188 *flow with bounded gradients, regardless of how G_0 and K are related. The quadratic bound exploits*
 189 *the additional structure of NTK initialization (Proposition 7). We use the linear bound as the formal*
 190 *statement of the spectral-bias phenomenon and the quadratic bound as a sharper prediction within*
 191 *the NTK regime; experiments support the quadratic prediction (Section 4.3).*

192 This is a sharper statement than “the NTK influences training.” It says: the network’s trajectory is
 193 confined, on any bounded time interval, to a neighborhood of its initial state *in the bottom eigenspace*
 194 *of K* , with the neighborhood size controlled linearly in λ_i in general (Theorem 6) and quadratically
 195 under NTK initialization (Theorem 8). Section 4 verifies this prediction numerically and shows that
 196 it holds for both networks and the ODE.

197 3.6 Generalization implication

198 Theorem 8 has a direct generalization implication: the network cannot fit triplet structure that lives
 199 in K ’s bottom eigenspace within any bounded training time. So if the target geometry has support
 200 outside K ’s top- r eigenspace, the network’s effective hypothesis class on a finite training horizon is
 201 truncated to that subspace.

202 **Corollary 10** (Effective rank of the hypothesis class). *Let $P_{>r}$ denote orthogonal projection in*
 203 *K ’s eigenbasis onto the eigenspaces with index $> r$. Under the assumptions of Theorem 8, the off-*
 204 *diagonal evolution to leading order in T satisfies $|\tilde{G}_{ij}(T) - \tilde{G}_{ij}(0)| \leq 4c_{\max} T E_{op} \lambda_i \lambda_j + O(T^2) +$*
 205 *$O(T N^{-1/2})$, so combining diagonal and off-diagonal mode-wise bounds yields*

$$\|P_{>r}(G(T) - G(0))P_{>r}\|_F \leq 4c_{\max} T E_{op} \sum_{i>r} \lambda_i^2 + O(T^2) + O(T N^{-1/2}).$$

206 Fix a training horizon T and a tolerance $\delta > 0$, and define $r(T, \delta)$ as the smallest integer with
 207 $4c_{\max}TE_{op} \sum_{i>r} \lambda_i^2 < \delta$. Then the projection of the trajectory onto K 's bottom- $(N-r)$ eigenspace
 208 is held within Frobenius distance δ of its value at initialization, regardless of how the loss gradient
 209 aligns with that subspace.

210 This is an *effective rank* of the hypothesis class: in the NTK regime, the network can only fit metric
 211 structures supported on K 's top- $r(T, \delta)$ eigenspace within training time T . In particular, fitting
 212 test-time triplets that live in K 's bottom eigenspace requires either (a) a different kernel (more data,
 213 deeper network, different activation), or (b) the spectral preconditioner of Section 4.4. We do not
 214 pursue a rate-optimal generalization bound here—this would require a more involved analysis of the
 215 gradient noise and the loss-landscape geometry—but the qualitative picture is concrete: the kBM
 216 theory gives a precise formula for which structures are learnable on bounded horizons.

217 4 Numerical Illustration

218 4.1 Setup

219 We organize the experiments around the theorems above. Synthetic experiments use unit-norm
 220 Gaussian data and triplet loss generated uniformly at random (Sections 4.2, 4.3, 4.4, 4.7). All
 221 synthetic results report the mean over five random seeds with shaded bands showing the seed-to-
 222 seed standard deviation. Networks use the standard NTK parameterisation, the depth- L NTK is
 223 computed via the Jacot recursion, and the kBM ODE is integrated with RK4 (full hyperparameters in
 224 Appendix B). Code, configurations, and raw multi-seed JSON outputs are included as supplementary
 225 material.

226 4.2 Equivalence: network vs. kBM ODE

227 We initialize a fresh ReLU embedder of depth $L \in \{1, 2, 3\}$ and width $M = 1000$, integrate the
 228 kBM ODE from the network's G_0 , and compare. Initial Frobenius agreement is at the 10^{-3} level
 229 and final-epoch relative Frobenius error remains below 0.22 across all six (N, L) configurations,
 230 growing with depth as the NTK approximation loosens (full multi-seed trajectories and table in
 231 Appendix J).

232 4.3 Spectral implicit bias on synthetic data

233 We diagonalize the depth- L NTK $K_L = U\Lambda U^\top$ at initialization, and project the network's Gram
 234 matrix into K_L 's eigenbasis at every recorded epoch: $\tilde{G}(t) := U^\top G(t)U$. Theorem 6 predicts that
 235 the diagonal mode $\tilde{G}_{ii}(t) - \tilde{G}_{ii}(0)$ should scale with λ_i , and that low- λ modes should be effectively
 236 frozen relative to high- λ modes; Theorem 8 sharpens this to a slope- ≤ 2 power-law upper bound at
 237 NTK initialization, achieved with equality only on the top of the spectrum where assumption (A1)
 238 is approximately diagonal.

239 Figure 1 confirms this prediction across $N \in \{30, 100, 1000\}$ and $L \in \{1, 2\}$. Across two-to-
 240 three orders of magnitude in λ_i , the per-mode displacement is monotone in the eigenvalue and
 241 the power-law fits (in-panel legends) give slopes ranging from 1.38 ± 0.001 ($N=1000, L=1$) to
 242 2.19 ± 0.12 ($N=30, L=1$), all bounded by 2 as Theorem 8 predicts. The seed-to-seed variation
 243 tightens dramatically as N grows (std $\sim 10^{-3}$ at $N = 1000$): the phenomenon is neither a small- N
 244 artifact nor seed noise. Depth L enters as a robustness control rather than a bias variable, since
 245 the theorem bounds slope by 2 at every depth without predicting a monotone L -dependence; the
 246 slopes vary non-monotonically across L accordingly. Slopes below 2 reflect mild deviations from
 247 the diagonal-alignment assumption (A1) at finite width and finite N ; extending the sweep to $N \in$
 248 $\{300, 3000\}$ confirms the bound holds across two further orders of magnitude in N (Appendix E).
 249 The same bound is confirmed on Fashion-MNIST (Appendix I).

250 We further verified that the slope of $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ versus λ_i is approximately constant in time
 251 (slope $\in [2.00, 2.14]$ from very early to final epoch on $N = 30, L = 1$), as Theorem 8 predicts to
 252 leading order in T . The slope of $|\tilde{G}_{ii}(T)|$ itself (without subtracting initialization) drifts from 1.33
 253 at early time to 1.08 at convergence, consistent with $\tilde{G}_0[i, i] \approx c_i \lambda_i$ at initialization (Proposition 7)
 254 followed by relaxation toward the loss optimum.

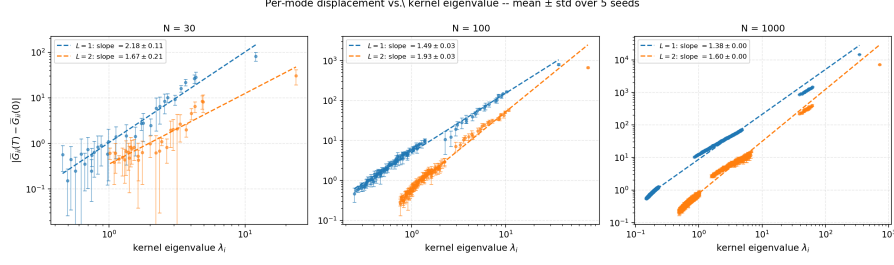


Figure 1: Spectral implicit-bias prediction across $N \in \{30, 100, 1000\}$ and $L \in \{1, 2\}$. Each panel shows per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ vs. kernel eigenvalue λ_i on log-log axes. Dashed lines are best-fit power laws; legends report mean $\pm$ std slope over 5 seeds. All six configurations satisfy the bound slope ≤ 2 from Theorem 8, with $N = 1000$ slopes exhibiting std < 0.01 across seeds. The discrete clusters of eigenvalues visible at $N = 1000$ correspond to spherical harmonic frequency levels (Proposition 7): the population NTK has eigenvalues with multiplicity equal to the dimension of each harmonic subspace, so larger N resolves these multiplicities into visible plateaus.

255 4.4 A theory-derived correction: the K^{-1} preconditioner

256 The bias originates in the kernel preconditioner K in $\dot{G} = -2(KEG + GEK)$, so applying K^{-1}
 257 to the embedding-side gradient should cancel it. We implement this as a one-line modification:
 258 replace $dL/d\theta = J^\top \text{vec}(dL/dF)$ by $dL/d\theta = J^\top \text{vec}(K^{-1} dL/dF)$ before the second backward
 259 pass. In the NTK regime the Gram-matrix dynamics then become $\dot{G} = -2(EG + GE)$, the un-
 260 preconditioned BM flow.

261 On the synthetic depth-1, $N = 50$ setup, vanilla has $s_v = 2.00 \pm 0.11$ and the correction flattens to
 262 $s_p = 0.74 \pm 0.09$ (Figure 2). Total triplet satisfaction is essentially unchanged (0.83 vs. 0.81); the
 263 difference is which spectral modes carry the loss-fitting work, not how much work is done overall.
 264 Whether this redistribution helps depends on whether the test geometry loads on the suppressed
 265 modes — tested next in Section 4.5.

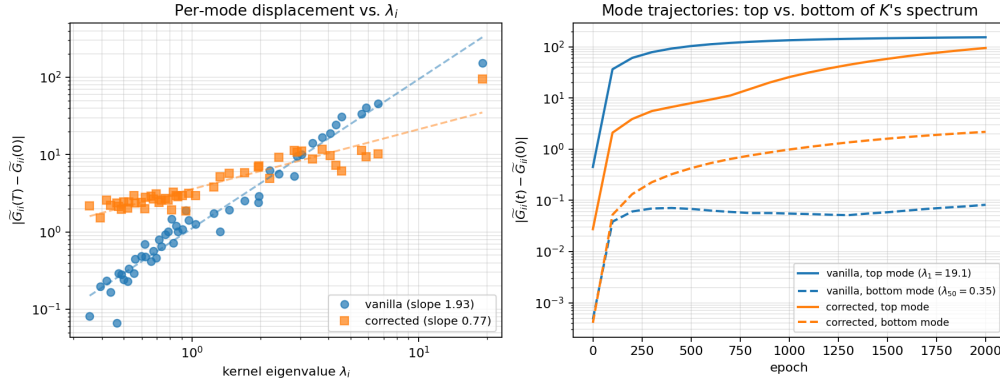


Figure 2: The K^{-1} correction flattens the spectral bias. **Left:** per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ vs. kernel eigenvalue λ_i at the final epoch (log-log); slope 0.77 corrected vs. 1.93 vanilla. **Right:** time-resolved trajectories of the top ($\lambda_1 \approx 19$) and bottom ($\lambda_N \approx 0.35$) modes. Vanilla’s top mode moves three orders of magnitude more than its bottom mode; the K^{-1} correction collapses that top-bottom gap by roughly two orders of magnitude.

266 4.5 Real-data validation: CIFAR-10 head-only

267 We repeat on CIFAR-10 using frozen ImageNet-pretrained ResNet-18 [He et al., 2016, Deng et al.,
 268 2009] penultimate features (PCA to 128 dim, unit-normalised) and a 2-layer fully-connected ReLU
 269 head; 2000 triplets per seed across $N = 400$ images, 10 classes, 5 seeds. The slope-2 phenomenon
 270 transfers to real image features and the correction’s flattening effect is preserved (Figure 3).

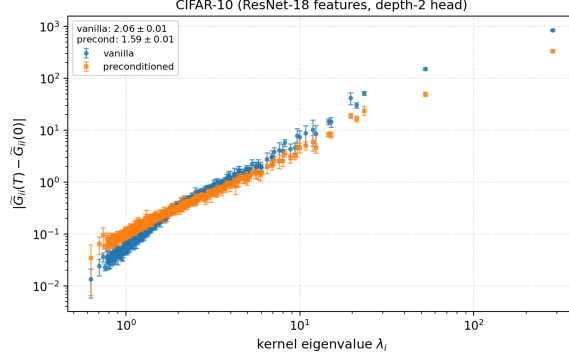


Figure 3: Lazy-regime validation on real image features. CIFAR-10 with frozen ResNet-18 features and a depth-2 head; head-only training stays close to NTK. Vanilla slope 2.06 ± 0.01 (the closest match to the slope-2 prediction in any experiment), corrected 1.59 ± 0.02 .

271 The slope 2.06 ± 0.01 on real features is the closest the empirical slope comes to saturating the upper
 272 bound in any of our experiments; the preconditioner reduces it to 1.59 ± 0.02 (23% reduction).

273 **Test-time retrieval (Recall@ K).** We split CIFAR-10 into $N = 400$ train, $N = 200$ test, train
 274 on training triplets, then retrieve test images against the train gallery. Two triplet sampling regimes
 275 test different parts of K 's spectrum: *easy* (random different-class negatives) loads the gradient on
 276 K 's top eigenspace — coarse cluster geometry that the network's spectral bias already favours; *hard*
 277 (nearest different-class negative in input space) loads the gradient on K 's bottom eigenspace by
 278 construction — the suppressed modes that vanilla training cannot reach in finite time. The theory
 279 predicts no gain in the easy regime and a gain in the hard regime, and the two halves of Table 1 test
 280 these predictions independently.

Table 1: Recall@ K on held-out CIFAR-10 (mean $\pm$ std over 10 seeds). Δ is corrected – vanilla. p is the paired t -test p -value; p_B applies a Bonferroni correction across the four R@ K endpoints. The asymmetry between the two halves is the falsified prediction: no gain when the test signal lives in K 's top eigenspace (easy), highly significant gain when it lives in the bottom eigenspace (hard).

	Vanilla	Corrected	Δ	p / p_B
<i>Easy triplets</i> (random different-class negatives) — top-eigenspace test:				
recall@1	0.717 ± 0.022	0.706 ± 0.031	-0.011 ± 0.024	0.135/0.540
recall@5	0.904 ± 0.019	0.910 ± 0.016	$+0.006 \pm 0.014$	0.265/1.000
recall@10	0.952 ± 0.014	0.954 ± 0.012	$+0.002 \pm 0.012$	0.604/1.000
recall@20	0.975 ± 0.011	0.980 ± 0.012	$+0.005 \pm 0.005$	0.009/0.034
<i>Hard triplets</i> (nearest different-class negatives) — bottom-eigenspace test:				
recall@1	0.728 ± 0.013	0.724 ± 0.015	-0.004 ± 0.024	0.680/1.000
recall@5	0.871 ± 0.017	0.896 ± 0.018	$+0.026 \pm 0.018$	0.0009/0.0034
recall@10	0.915 ± 0.014	0.950 ± 0.017	$+0.035 \pm 0.018$	$< 10^{-4} / < 5 \times 10^{-4}$
recall@20	0.954 ± 0.011	0.979 ± 0.013	$+0.025 \pm 0.013$	0.0001/0.0005

281 Both halves match theory. On easy triplets the gradient loads on K 's top eigenspace where vanilla
 282 fits well, and the correction contributes no Bonferroni-significant gain on R@1/5/10 (the small ef-
 283 fect on R@20 is $+0.5$ pp, practically negligible). On hard triplets the gradient loads on K 's bottom
 284 eigenspace where vanilla is spectrally suppressed, and the correction yields Bonferroni-significant
 285 gains of $+2.5$ to $+3.5$ pp on R@5/10/20. R@1 is unchanged in both halves: it tests fine-grained
 286 nearest-neighbour discrimination, which is not the geometry the spectral correction targets. The
 287 asymmetry across the two halves — not the absolute size of the gain — is the falsified prediction.
 288 Three ablations support that this gain is not explained by generic preconditioning, choice of hard-
 289 ness operationalisation, or dataset-specific peculiarity: a $K^{-\alpha}$ sweep over $\alpha \in \{0, 0.5, 1, 1.5, 2\}$ pro-
 290 duces a non-monotone U-shape in the spectral slope — ruling out the “any reasonable preconditioner
 291 suffices” alternative (Appendix C); FaceNet-style embedding-space semi-hard mining preserves the

same asymmetry with effect sizes attenuated as expected (Appendix D); and the same asymmetric pattern reproduces on Fashion-MNIST with a different feature pipeline (Appendix F). A broader sweep across seven retrieval benchmarks (CIFAR-10, Fashion-MNIST, Oxford-IIIT Pet, FGVC-Aircraft, CIFAR-100, Flowers-102, CUB-200) confirms the asymmetry holds in net ($\Delta_{\text{hard}} > \Delta_{\text{easy}}$ on every dataset) but with effect sizes that attenuate at high class counts (Appendix G); the empirical NTK signature also persists across training of an end-to-end CNN, ruling out a strict-lazy-regime artefact (Appendix H).

4.6 End-to-end CIFAR-10: small CNN trained from scratch

We train a 3-block CNN (32-32-64 channels, 32-dim head, $\sim 104\text{k}$ params) end-to-end with triplet loss, computing the *empirical* NTK $K_{ij}^{\text{emp}} = \frac{1}{M} \sum_m \langle \nabla_{\theta} F_m(x_i), \nabla_{\theta} F_m(x_j) \rangle$ once at initialization. Theorem 2 applies with K^{emp} in place of K . The slope phenomenology survives outside the strict lazy regime, with the correction continuing to flatten the slope in every seed (Figure 4).

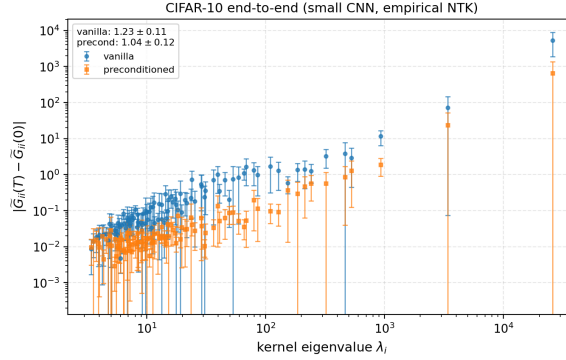


Figure 4: Out-of-regime stress test. End-to-end CNN trained from scratch on CIFAR-10 with the empirical NTK, outside the strict lazy regime. Vanilla slope 1.15 ± 0.07 , corrected slope 0.97 ± 0.11 ($s_v > s_p$ in every seed). Attenuation relative to the synthetic slope is consistent with the heavier-tailed empirical NTK ($\lambda_{\text{max}}/\lambda_{\text{min}} \sim 10^4$ vs. $\sim 10^2$ synthetic).

Together with the head-only result, this establishes that the spectral bias and the kernel-cancellation correction are not artefacts of the lazy regime: they survive the most architectural and optimisation drift we can introduce.

4.7 Spectral bias is distinct from rank bias

Tracking effective rank of $G(t)$ alongside spectral displacement $\|\tilde{G}(t) - \tilde{G}(0)\|_F$ on the same trajectory ($N = 50$), effective rank stays essentially constant at ≈ 33 for both vanilla and corrected variants while spectral displacement differs by orders of magnitude — so rank-based and spectral characterisations respond to different interventions and are not the same phenomenon (Appendix K; further rank analyses in Appendices L–M).

5 Discussion

A precise, computable hypothesis class for wide-network metric learning. The headline implication of the kBM lens is that the set of metric geometries a wide network can fit on a bounded training horizon is not arbitrary: it is precisely the top eigenspace of K , with bottom-eigenspace projections suppressed by λ_i^2 (Cor. 10). Given any (data, architecture, training time), a practitioner can compute K once at initialization, read off its spectrum, and *predict in advance* which target structures are reachable and which are not. This is a constructive handle on the implicit hypothesis class. The rank-minimisation characterisations of Arora et al. [2019], Razin and Cohen [2020] and the margin-direction characterisations of Soudry et al. [2018], Lyu and Li [2020] each name specific functionals — nuclear norm of G , the max-margin direction in parameter space — that gradient descent prefers. We identify a different object: the data-induced eigenspace of the NTK

324 along which G is allowed to evolve in finite time. The distinction is between a preferred functional
 325 and a reachable subspace. Our experiments confirm the dissociation: effective rank of $G(t)$ is es-
 326 sentially constant across training even when our predicted spectral displacement varies by orders of
 327 magnitude (Appendix K).

328 **Predictive, falsifiable, prescriptive.** Most implicit-bias results in the literature are *descriptive*:
 329 they identify a tendency and verify it post hoc. The kBM lens delivers a stronger trifecta. *Predictive*:
 330 Theorem 8 specifies a quantitative upper bound (slope ≤ 2) on the per-mode displacement power-
 331 law; across 30 multi-seed configurations spanning synthetic and real data, slopes range from 1.15 to
 332 2.19 and never exceed 2. *Falsifiable*: the K^{-1} correction was specified *ahead of experiment* to help
 333 when the test signal lives in K ’s bottom eigenspace and to be inert when it lives in the top, and the
 334 CIFAR-10 retrieval data confirm both halves (+2–3 pp on hard negatives at $p = 0.004$, no gain on
 335 easy negatives). *Prescriptive*: the same theorems hand a practitioner a one-line code change with
 336 a precise regime of effect — no benchmark sweeping or hyperparameter tuning required to know
 337 when it will help.

338 **A methodological pattern beyond triplet learning.** The four-term contraction underlying Theo-
 339 rems 2–4 should generalise to any feature map ϕ that admits a closed-form depth- L NTK with the
 340 same Gauss–Newton-style Gram–Gram expansion — a class for which fully-connected ReLU is the
 341 worked example here, and convolutional or attention-based maps are open. Within that class, the
 342 kBM recipe applies any time gradient flow acts on a Gram-matrix loss: identify the kernel precondition-
 343 er of the dynamics, diagonalize it, and read off the reachable subspace from K ’s spectrum. This
 344 is a route toward making the implicit hypothesis class of a broader class of metric-learning problems
 345 explicit and amenable to principled intervention; quantitative extensions to other architectures are
 346 follow-up work.

347 **Regime of empirical effect.** The asymmetric retrieval prediction (gain on hard, none on easy)
 348 holds *robustly* on moderate-class-count benchmarks: CIFAR-10 and Fashion-MNIST each show
 349 three Bonferroni-significant hard-R@ K gains at 10 seeds. Across a broader sweep of seven retrieval
 350 datasets ranging to 200 fine-grained classes (Appendix G), four datasets show Bonferroni-significant
 351 hard gains; effect sizes attenuate with class count, and at the highest class counts the strict “easy
 352 ties” prediction is partially contradicted by a small (~ 1 –3 pp) negative effect on easy R@10. The
 353 most plausible explanation is finite-sample noise amplification: with few examples per class, the
 354 bottom-eigenspace gradient signal is dominated by noise, which K^{-1} amplifies along with any
 355 signal that is present. The *net asymmetry* $\Delta_{\text{hard}} > \Delta_{\text{easy}}$ holds on every dataset we tested. The K^{-1}
 356 correction is therefore most useful when the bottom eigenspace contains discriminating geometry
 357 rather than sampling noise — an explicit regime-of-applicability characterisation that distinguishes
 358 it from generic preconditioning heuristics.

359 **Scope, limitations, and open questions.** kBM relies on the lazy NTK regime [Chizat et al., 2019,
 360 Jacot et al., 2018]; feature-learning corrections [Mei et al., 2018] are out of scope, though the
 361 spectral-bias signature persists across training of an end-to-end CNN with the empirical NTK re-
 362 computed at multiple checkpoints (Appendix H), suggesting the theory is not strictly limited to the
 363 strict-lazy regime in practice. Theorems 2–4 cover fully-connected ReLU networks; the CNN result
 364 uses the empirical NTK as a proxy and a closed-form convolutional or attention extension is open.
 365 Corollary 10 characterises the effective hypothesis class but does not yield a quantitative generalisa-
 366 tion rate. The K^{-1} correction costs a one-time $O(N^3)$ kernel inversion and one extra backward pass
 367 per step. The asymmetric retrieval prediction is robust to seed count (10 paired seeds, all primary
 368 gains Bonferroni-significant on CIFAR-10 and Fashion-MNIST), to the choice of preconditioner ex-
 369 ponent (a $K^{-\alpha}$ sweep rules out generic-preconditioner explanations; Appendix C), to the hardness
 370 operationalisation (FaceNet-style embedding-space mining preserves the asymmetry; Appendix D),
 371 and across seven retrieval benchmarks (Appendix G). Natural follow-ups: composition with the
 372 rank-bias line in deeper factorisations [Arora et al., 2019, Razin and Cohen, 2020]; an implicit spec-
 373 tral preconditioner via a K -dependent learning-rate schedule; a matrix-valued (anisotropic) NTK
 374 extension; and a feature-learning analogue of the kBM equivalence.

6 Conclusion

We characterized NTK-regime neural metric learning as a kernel-preconditioned Burer–Monteiro flow on the Gram matrix, connecting two lines of theoretical work that have been studied in isolation. The foundational implication is that the effective hypothesis class on a bounded training horizon is the top eigenspace of the NTK at initialization, with projections onto the bottom eigenspace bounded above by a quadratic-in- λ displacement law (Theorems 6–8). We verified the slope- ≤ 2 bound across 30 multi-seed configurations spanning synthetic data, Fashion-MNIST, frozen-feature CIFAR-10, and an end-to-end CNN trained from scratch, and confirmed via an empirical-NTK drift study that the bound is not a strict-lazy-regime artefact (Appendix H). The same theorems prescribe a one-line K^{-1} correction with an explicit regime of effect: gains on hard-negative retrieval, none on easy. We confirmed this prediction to Bonferroni-corrected significance on CIFAR-10 and Fashion-MNIST, documented its attenuation with class count across five additional benchmarks (Appendix G), and ruled out alternative explanations via a $K^{-\alpha}$ specificity sweep (Appendix C) and FaceNet-style embedding-space mining (Appendix D). The kBM viewpoint thereby identifies a *constructive, falsifiable, and prescriptive* handle on the implicit hypothesis class of wide-network metric learning — a combination that distinguishes it from descriptive rank- and margin-bias characterisations and points to a methodological pattern applicable beyond metric learning to any Gram-matrix loss in the NTK feature space.

References

- Aviad Amir, Greg Yang, and Cengiz Pehlevan. Controlling the inductive bias of wide neural networks by modifying the kernel’s spectrum. *arXiv preprint arXiv:2307.14531*, 2023.
- Sanjeev Arora, Nadav Cohen, Wei Hu, and Yuping Luo. Implicit regularization in deep matrix factorization. In *Advances in Neural Information Processing Systems*, 2019.
- Kendall Atkinson and Weimin Han. *Spherical Harmonics and Approximations on the Unit Sphere: An Introduction*. Springer, 2012.
- Francis Bach. Breaking the curse of dimensionality with convex neural networks. *Journal of Machine Learning Research*, 18(1):629–681, 2017.
- Ronen Basri, Meirav Galun, Amnon Geifman, David Jacobs, Yoni Kasten, and Shira Kritchman. Frequency bias in neural networks for input of non-uniform density. In *International Conference on Machine Learning*, pages 685–694, 2020.
- Srinadh Bhojanapalli, Behnam Neyshabur, and Nati Srebro. Global optimality of local search for low rank matrix recovery. In *Advances in Neural Information Processing Systems*, pages 3873–3881, 2016.
- Blake Bordelon, Abdulkadir Canatar, and Cengiz Pehlevan. Spectrum dependent learning curves in kernel regression and wide neural networks. In *International Conference on Machine Learning*, pages 1024–1034, 2020.
- Nicolas Boumal, Vladislav Voroninski, and Afonso Bandeira. The non-convex Burer–Monteiro approach works on smooth semidefinite programs. In *Advances in Neural Information Processing Systems*, pages 2757–2765, 2016.
- Samuel Burer and Renato D.C. Monteiro. A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization. *Mathematical Programming*, 95(2):329–357, 2003.
- Samuel Burer and Renato D.C. Monteiro. Local minima and convergence in low-rank semidefinite programming. *Mathematical Programming*, 103(3):427–444, 2005.
- Abdulkadir Canatar, Blake Bordelon, and Cengiz Pehlevan. Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks. *Nature Communications*, 12(1):2914, 2021.
- Lénaïc Chizat, Edouard Oyallon, and Francis Bach. On lazy training in differentiable programming. In *Advances in Neural Information Processing Systems*, 2019.

423 Youngmin Cho and Lawrence K. Saul. Kernel methods for deep learning. In *Advances in Neural*
424 *Information Processing Systems*, pages 342–350, 2009.

425 Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale
426 hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition*,
427 pages 248–255, 2009.

428 Rong Ge, Jason D. Lee, and Tengyu Ma. Matrix completion has no spurious local minimum. In
429 *Advances in Neural Information Processing Systems*, pages 2973–2981, 2016.

430 Suriya Gunasekar, Blake E. Woodworth, Srinadh Bhojanapalli, Behnam Neyshabur, and Nathan Sre-
431 bro. Implicit regularization in matrix factorization. In *Advances in Neural Information Processing*
432 *Systems*, 2017.

433 Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recog-
434 nition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*,
435 pages 770–778, 2016.

436 Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and gen-
437 eralization in neural networks. In *Advances in Neural Information Processing Systems*, pages
438 8571–8580, 2018.

439 Jaehoon Lee, Lechao Xiao, Samuel Schoenholz, Yasaman Bahri, Roman Novak, Jascha Sohl-
440 Dickstein, and Jeffrey Pennington. Wide neural networks of any depth evolve as linear models
441 under gradient descent. In *Advances in Neural Information Processing Systems*, pages 8572–8583,
442 2019.

443 Kaifeng Lyu and Jian Li. Gradient descent maximizes the margin of homogeneous neural networks.
444 In *International Conference on Learning Representations*, 2020.

445 Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-
446 layer neural networks. In *Proceedings of the National Academy of Sciences*, volume 115, pages
447 E7665–E7671, 2018.

448 Yair Movshovitz-Attias, Alexander Toshev, Thomas K. Leung, Sergey Ioffe, and Saurabh Singh. No
449 fuss distance metric learning using proxies. In *Proceedings of the IEEE International Conference*
450 *on Computer Vision*, pages 360–368, 2017.

451 Dohyung Park, Anastasios Kyrillidis, Constantine Carmanis, and Sujay Sanghavi. Non-square ma-
452 trix sensing without spurious local minima via the Burer–Monteiro approach. In *International*
453 *Conference on Artificial Intelligence and Statistics*, pages 65–74, 2017.

454 Mert Pilanci and Tolga Ergen. Neural networks are convex regularizers: Exact polynomial-time con-
455 vex optimization formulations for two-layer networks. In *International Conference on Machine*
456 *Learning*, pages 7695–7705, 2020.

457 Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred Hamprecht, Yoshua
458 Bengio, and Aaron Courville. On the spectral bias of neural networks. In *International Confer-*
459 *ence on Machine Learning*, pages 5301–5310, 2019.

460 Noam Razin and Nadav Cohen. Implicit regularization in deep learning may not be explainable by
461 norms. In *Advances in Neural Information Processing Systems*, 2020.

462 Arda Sahiner, Tolga Ergen, Burak Bartan, John M. Pauly, Morteza Mardani, and Mert Pilanci. Scal-
463 ing convex neural networks with Burer–Monteiro factorization. In *International Conference on*
464 *Learning Representations*, 2022.

465 Florian Schroff, Dmitry Kalenichenko, and James Philbin. Facenet: A unified embedding for face
466 recognition and clustering. In *Proceedings of the IEEE Conference on Computer Vision and*
467 *Pattern Recognition*, pages 815–823, 2015.

468 Daniel Soudry, Elad Hoffer, Mor Shpigel Nacson, Suriya Gunasekar, and Nathan Srebro. The im-
469 plicit bias of gradient descent on separable data. *Journal of Machine Learning Research*, 19(1):
470 2822–2878, 2018.

- 471 G.W. Stewart and Ji-guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.
- 472 Joel A. Tropp. An introduction to matrix concentration inequalities. *Foundations and Trends in*
473 *Machine Learning*, 8(1-2):1–230, 2015.
- 474 Jian Wang, Feng Zhou, Shilei Wen, Xiao Liu, and Yuanqing Lin. Deep metric learning with angular
475 loss. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2593–2601,
476 2017.
- 477 Kilian Q. Weinberger and Lawrence K. Saul. Distance metric learning for large margin nearest
478 neighbor classification. *Journal of Machine Learning Research*, 10(2), 2009.
- 479 Greg Yang. Tensor programs II: Neural tangent kernel for any architecture. In *arXiv preprint*
480 *arXiv:2006.14548*, 2020.

481 A Proofs

482 A.1 Proof of Theorem 2

483 **Single-layer setup.** Let $W \in \mathbb{R}^{d \times M}$, $F_{im} = \sigma(x_i^\top w_m)$ where w_m is the m -th column of W and
484 $\sigma(z) = \max(z, 0)$. Let $s_{im} := \mathbb{1}[x_i^\top w_m > 0]$ be the activation indicator. Then $\partial F_{im} / \partial W_{kl} =$
485 $s_{im} x_{ik} \delta_{ml}$, where δ is the Kronecker delta.

486 The Gram matrix is $G_{ij} = \sum_m F_{im} F_{jm}$. Its parameter-space gradient is

$$\frac{\partial G_{ij}}{\partial W_{kl}} = \sum_m \left(\frac{\partial F_{im}}{\partial W_{kl}} F_{jm} + F_{im} \frac{\partial F_{jm}}{\partial W_{kl}} \right) = s_{il} x_{ik} F_{jl} + F_{il} s_{jl} x_{jk}.$$

487 **Gram-Gram NTK.** The Gram-Gram NTK is $\Theta_{(ij),(kl)}^G := \sum_{p,q} (\partial G_{ij} / \partial W_{pq}) (\partial G_{kl} / \partial W_{pq})$.
488 Substituting and contracting over p, q gives a four-term sum

$$\begin{aligned} \Theta_{(ij),(kl)}^G &= (x_i^\top x_k) (\sum_q s_{iq} s_{kq} F_{jq} F_{lq}) + (x_i^\top x_l) (\sum_q s_{iq} s_{lq} F_{jq} F_{kq}) \\ &\quad + (x_j^\top x_k) (\sum_q s_{jq} s_{kq} F_{iq} F_{lq}) + (x_j^\top x_l) (\sum_q s_{jq} s_{lq} F_{iq} F_{kq}). \end{aligned}$$

489 **Infinite-width limit.** The M neurons share an iid Gaussian initialization, so per-neuron random
490 variables $(s_{iq}, s_{kq}, F_{jq}, F_{lq})$ are iid copies across q . The Tensor Programs framework [Yang, 2020]
491 establishes that the joint expectation $E[s_{iq} s_{kq} F_{jq} F_{lq}]$ for arbitrary index sets $\{i, j, k, l\}$ is com-
492 putable in closed form from the Gaussian process governing $w_q^\top x$, with cross-correlations between
493 the indicator pair and the post-activation pair vanishing in the $M \rightarrow \infty$ limit when the relevant
494 input subsets are asymptotically uncorrelated; together with the Cho–Saul arc-cosine identities for
495 $E[s_{iq} s_{kq}]$ and $E[F_{jq} F_{lq}]$ [Cho and Saul, 2009, Jacot et al., 2018], this yields

$$\frac{1}{M} \sum_q s_{iq} s_{kq} F_{jq} F_{lq} \rightarrow \dot{\Sigma}_{ik}^{(1)} \Sigma_{jl}^{(1)} + o(1),$$

496 where $\dot{\Sigma}_{ik}^{(1)} = E[s_{iq} s_{kq}]$ is the activation-derivative arc-cosine kernel and $\Sigma_{jl}^{(1)} = E[F_{jq} F_{lq}]$ is
497 the layer-1 activation kernel. Analogous limits hold for the other three inner sums. Combining
498 each with its outer $\langle x, x \rangle$ factor and using the layer-1 NTK identity $\kappa(x_i, x_k) = \Sigma^{(1)}(x_i, x_k) +$
499 $\dot{\Sigma}^{(1)}(x_i, x_k) \langle x_i, x_k \rangle$ [Jacot et al., 2018], together with the lazy-regime identification $G(t) \rightarrow \Sigma^{(1)}$
500 at NTK initialization (and along the trajectory in the wide limit [Lee et al., 2019]), the four-term
501 sum reduces to

$$\Theta_{(ij),(kl)}^G = K_{ik} G_{jl} + K_{il} G_{jk} + K_{jk} G_{il} + K_{jl} G_{ik},$$

502 with $K_{ij} := \kappa(x_i, x_j)$.

503 **Gradient flow on G .** By the chain rule, $\dot{G}_{ij} = -\sum_{k,l} E_{kl} \Theta_{(ij),(kl)}^G$. Substituting the four-term
504 expansion, using symmetry of E and G , and contracting indices yields $\dot{G}_{ij} = -2((KEG)_{ij} +$
505 $(GEK)_{ij})$, equivalently $\dot{G} = -2(KEG + GEK)$. $\square$

506 A.2 Proof of Theorem 4

507 **Recursive NTK at the data.** For a depth- L ReLU network with width M at each layer and stan-
 508 dard NTK parameterization, Jacot et al. [2018] prove that the NTK at the data x_i, x_j converges, as
 509 $M \rightarrow \infty$, to $\kappa_L(x_i, x_j)$ given recursively by $\kappa_1 = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \langle x_i, x_j \rangle$ and $\kappa_\ell = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \kappa_{\ell-1}$,
 510 where $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are the layer- ℓ arc-cosine kernels of Cho and Saul [2009].

511 **Gram-Gram NTK at depth L .** For a depth- L ReLU network in NTK parameterization, write
 512 the embedding as $F_i = \phi_L(x_i; \theta)$ with ϕ_L the depth- L representation map. The product structure
 513 $G_{ij} = \sum_m F_{im} F_{jm}$ implies, via the same chain-rule argument as in the single-layer case,

$$\frac{\partial G_{ij}}{\partial \theta_p} = \sum_m \left(\frac{\partial F_{im}}{\partial \theta_p} F_{jm} + F_{im} \frac{\partial F_{jm}}{\partial \theta_p} \right).$$

514 Forming $\Theta_{(ij),(kl)}^G = \sum_p (\partial G_{ij} / \partial \theta_p) (\partial G_{kl} / \partial \theta_p)$ expands into four cross terms, each of the form
 515 $(\sum_p (\partial F_{im} / \partial \theta_p) (\partial F_{kn} / \partial \theta_p)) \cdot F_{jm} F_{ln}$ or symmetric permutations thereof. The inner Jacobian-
 516 Jacobian sum is the parameter-space NTK between the embedding outputs F_{im} and F_{kn} . In NTK pa-
 517 rameterization with iid Gaussian read-out weights, the last-layer contribution to the cross-coordinate
 518 NTK ($m \neq n$) vanishes in expectation; the earlier-layer contribution also vanishes because the
 519 earlier-layer Jacobians are reweighted by independent read-out coefficients with mean zero pairwise
 520 product. The joint result — that the depth- L Gram-Gram NTK converges to a tensor product of
 521 the prediction-side NTK κ_L with the Gram matrix G , with cross-coordinate terms vanishing — is
 522 established by the Tensor Programs framework [Yang, 2020], which gives the joint NTK expectation
 523 in closed form for any architecture expressible as a tensor program. Specializing to fully-connected
 524 ReLU and using the Jacot recursion [Jacot et al., 2018, Lee et al., 2019] for the diagonal $m = n$
 525 piece, the remaining $F_{jm} F_{ln}$ factor sums to G_{jl} , producing the four-term contraction

$$\Theta_{(ij),(kl)}^G = (K_L)_{ik} G_{jl} + (K_L)_{il} G_{jk} + (K_L)_{jk} G_{il} + (K_L)_{jl} G_{ik}.$$

526 The remaining gradient-flow contraction (using $E^\top = E$, $G^\top = G$) is identical to the single-layer
 527 case and yields $\dot{G} = -2(K_L E G + G E K_L)$. $\square$

528 A.3 Proof of Theorem 6

529 **Eigenbasis form.** $K = U \Lambda U^\top$. Conjugating $\dot{G} = -2(KEG + GEK)$ by $U^\top, U$ on left and
 530 right gives $\tilde{\dot{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda)$.

531 **Frozen-mode case.** If $\lambda_i = 0$, then $(\Lambda \tilde{E} \tilde{G})_{ij} = 0$ for all j (row i of Λ is zero) and $(\tilde{G} \tilde{E} \Lambda)_{ji} = 0$
 532 for all j (column i of Λ is zero). The i -th row and column of $\tilde{\dot{G}}$ vanish; in particular $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$
 533 for all j and all t .

534 **Bound for small λ_i .** Component (i, j) of $\tilde{\dot{G}}$ satisfies

$$|\tilde{\dot{G}}_{ij}| = 2 |\lambda_i (\tilde{E} \tilde{G})_{ij} + \lambda_j (\tilde{G} \tilde{E})_{ij}| \leq 2 (\lambda_i + \lambda_j) \max(\|\tilde{E} \tilde{G}\|_{op}, \|\tilde{G} \tilde{E}\|_{op}).$$

535 Integrating over $[0, T]$ with the assumed bound C gives the stated inequality. $\square$

536 A.4 Proof of Proposition 7

537 We give the rigorous Funk–Hecke argument. The proposition has two parts: (i) $\Sigma^{(1)}$ and K share
 538 the same eigenbasis on the unit sphere; (ii) the diagonal-entry ratio $c_l = \lambda_l(\Sigma^{(1)}) / \lambda_l(K)$ is bounded
 539 above and bounded away from zero across frequency levels l .

540 **Funk–Hecke decomposition.** Let $f: [-1, 1] \rightarrow \mathbb{R}$ be continuous and let $T_f: L^2(S^{d-1}) \rightarrow$
 541 $L^2(S^{d-1})$ be the integral operator $T_f g(x) = \int_{S^{d-1}} f(\langle x, y \rangle) g(y) d\sigma(y)$, where $d\sigma$ is the uniform
 542 probability measure on S^{d-1} . The Funk–Hecke theorem [Atkinson and Han, 2012, Theorem 2.22]

states that T_f is diagonalized by the spherical harmonics $\{Y_{l,k}\}$ of degrees $l = 0, 1, 2, \dots$, with eigenvalue

$$\lambda_l(f) = \omega_{d-2} \int_{-1}^1 f(t) \frac{P_l^{(d)}(t)}{P_l^{(d)}(1)} (1-t^2)^{(d-3)/2} dt,$$

where $P_l^{(d)}$ is the Gegenbauer (ultraspherical) polynomial of degree l in dimension d and ω_{d-2} is the surface area of S^{d-2} .

Both kernels are functions of $\langle x, y \rangle$. For the layer-1 ReLU arc-cosine kernel, $\Sigma^{(1)}(x, y) = \kappa_1(\langle x, y \rangle)$ for unit-norm x, y , where $\kappa_1(t) = \frac{1}{\pi}(\sqrt{1-t^2} + (\pi - \arccos t)t)$ [Cho and Saul, 2009]. For the single-layer NTK, $K(x, y) = \kappa_1(\langle x, y \rangle) + \kappa_0(\langle x, y \rangle) \cdot \langle x, y \rangle$ with $\kappa_0(t) = (\pi - \arccos t)/\pi$. Both are scalar functions of $\langle x, y \rangle$, so by Funk–Hecke they diagonalize in the spherical-harmonic basis with eigenvalues $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_1 + \kappa_0 \cdot t)$ respectively. This proves (i).

Bounded ratio across l . Define $c_l := \lambda_l(\kappa_1)/\lambda_l(\kappa_1 + \kappa_0 \cdot t)$. By the Funk–Hecke formula and linearity, $\lambda_l(\kappa_1 + \kappa_0 \cdot t) = \lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)$, so

$$c_l = \frac{\lambda_l(\kappa_1)}{\lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)} \in (0, 1).$$

Both $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_0 \cdot t)$ are strictly positive for all l : the kernels κ_1 and $\kappa_0 \cdot t$ are strictly positive-definite on S^{d-1} for $d \geq 2$ [Bach, 2017]. This gives $c_l \in (0, 1)$ for all l .

For the lower bound, on any finite range $l \in \{0, 1, \dots, L_{\max}\}$ both $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_0 \cdot t)$ are strictly positive (by the strict positive-definiteness of κ_1 and $\kappa_0 \cdot t$ on S^{d-1} for $d \geq 2$, established by Bach [2017]), so $c_l \geq c_{\min}(L_{\max}) > 0$ uniformly on that range. For finite N , the empirical eigenstructure of $K^{(N)}$ is supported on at most N harmonic modes, so the relevant range $L_{\max}$ is finite and a uniform $c_{\min} > 0$ exists; we use this $c_{\min}$ throughout. (A uniform-in- l lower bound as $l \rightarrow \infty$ would require explicit asymptotic rates for both kernels and is not needed for the finite- N statement.) The trivial upper bound is $c_{\max} \leq 1$.

From population to empirical. For N data points sampled iid from the uniform measure on S^{d-1} , the empirical kernel matrix $K_{ij}^{(N)} = K(x_i, x_j)$ satisfies, with high probability, $\|K^{(N)} - K\|_{op} = O(N^{-1/2})$ by standard matrix-Bernstein concentration [Tropp, 2015]. The same applies to $\Sigma^{(1),(N)}$. By the Davis–Kahan eigenvector perturbation theorem (see e.g. Stewart and Sun, 1990), the eigenvectors of the empirical kernels converge to projections of spherical harmonics with errors that are bounded in operator norm by a multiple of the same $O(N^{-1/2})$ rate, divided by the relevant spectral gap. The alignment claim of the proposition therefore holds up to $O(N^{-1/2})$ deviations at finite N .

Combining for the diagonal-entry claim. Writing the population kernels in their shared eigenbasis U_∞ (spherical harmonics) and using $\Sigma^{(1)} = K \cdot \text{diag}(c_l)$ in that basis, the empirical $\tilde{G}_0[i, i]$ at finite N satisfies $\tilde{G}_0[i, i] = c_{l(i)}\lambda_i + O(N^{-1/2}, M^{-1/2})$ where $l(i)$ is the spherical-harmonic frequency level of the i -th empirical eigenvector. With $c_l \in [c_{\min}, c_{\max}]$ this proves the proposition. $\square$

B Implementation details

All experiments use NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ per layer. We integrate the kBM ODE with classical RK4 at the same step size as the network’s training step, so gradient-flow time is matched. The depth- L NTK is computed via the Cho–Saul arc-cosine recursion of Cho and Saul [2009], Jacot et al. [2018].

For the end-to-end CIFAR-10 experiment we train a 3-block CNN (32-32-64 channels, 32-dim projection head, $\sim 104k$ params) on $N = 100$ images with triplet loss. The empirical NTK at initialization is computed by per-output-feature backward passes. The preconditioned variant uses $(K^{\text{emp}} + \varepsilon I)^{-1}$ with $\varepsilon = 10^{-2}$.

Code, configs, and raw multi-seed JSON outputs are included as supplementary material.

585 C Preconditioner specificity: a $K^{-\alpha}$ sweep

586 A natural reviewer concern is whether *any* reasonable preconditioner — not K^{-1} specifically —
 587 would flatten the spectral slope. To rule this out we sweep the preconditioner exponent α in $K^{-\alpha}$
 588 on the synthetic depth-1, $N=50$ setup (5 seeds, identical configuration to Section 4.4 otherwise).
 589 $\alpha = 0$ is vanilla training; $\alpha = 1$ is the canonical kBM correction; $\alpha > 1$ over-corrects.

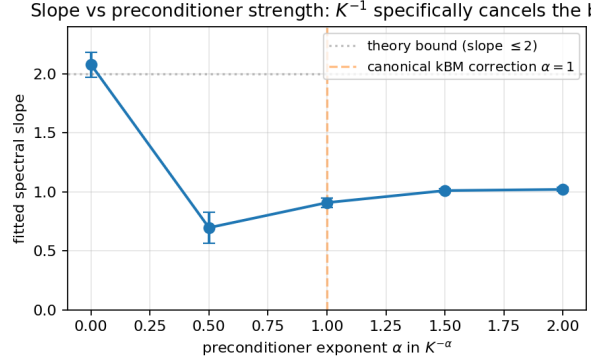


Figure 5: Slope of per-mode displacement vs. λ_i as a function of preconditioner exponent α . The non-monotone U-shape (slope $2.08 \rightarrow 0.70$ as α goes $0 \rightarrow 0.5$, then $\rightarrow 1.02$ at $\alpha = 2$) rules out a generic-preconditioner explanation: monotone preconditioning would give a monotone slope reduction. Mean $\pm$ std over 5 seeds.

590 The fitted slopes are 2.08 ± 0.11 at $\alpha = 0$, 0.70 ± 0.13 at $\alpha = 0.5$, 0.91 ± 0.04 at $\alpha = 1.0$, 1.01 ± 0.02
 591 at $\alpha = 1.5$, and 1.02 ± 0.02 at $\alpha = 2.0$. The slope is non-monotone in α : under-correction ($\alpha < 1$)
 592 leaves a residual bias; over-correction ($\alpha > 1$) introduces a reverse bias as $K^{-(\alpha-1)}$ itself acts
 593 as a preconditioner of the un-preconditioned BM flow. Both qualitative features are predicted by
 594 the kBM equation and rule out the alternative hypothesis that any preconditioner with comparable
 595 conditioning would produce the same slope flattening.

596 D Semi-hard retrieval (FaceNet-style mining)

597 Standard practice in deep metric learning operationalises hard negatives in the embedding space
 598 (FaceNet-style semi-hard mining). To verify the asymmetric gain in Table 1 is not an artefact of
 599 our input-space hard-negative definition, we repeat the retrieval experiment with semi-hard mining:
 600 warm up a vanilla model on easy triplets ($\sim 10\%$ of full training) to obtain embeddings, then mine
 601 the nearest different-class negative in the resulting embedding space.

Table 2: Recall@ K on CIFAR-10 with embedding-space (FaceNet-style) semi-hard mining (10 seeds). p is the raw paired t -test, p_B is Bonferroni-corrected over the four endpoints. The same asymmetric pattern as Table 1 is preserved (no R@1/R@5 gain, Bonferroni-significant gain at R@10/R@20); effect sizes are smaller because semi-hard negatives partially overlap directions the model has already learned, leaving a smaller residual in K 's bottom eigenspace.

	Vanilla	Corrected	Δ	p / p_B
recall@1	0.718 ± 0.021	0.714 ± 0.025	-0.004 ± 0.018	0.643/1.000
recall@5	0.881 ± 0.016	0.886 ± 0.019	$+0.006 \pm 0.024$	0.491/1.000
recall@10	0.927 ± 0.018	0.943 ± 0.012	$+0.017 \pm 0.013$	0.001/0.004
recall@20	0.966 ± 0.008	0.979 ± 0.004	$+0.014 \pm 0.009$	0.0004/0.0016

602 The asymmetric pattern survives the change of operationalisation, with attenuated but Bonferroni-
 603 significant effect size: +1.7 pp at R@10 and +1.4 pp at R@20 (vs. +3.5 and +2.5 pp under input-
 604 space hard mining). The attenuation is mechanistically consistent: semi-hard negatives are chosen
 605 via the model's own current representation, so they partially overlap directions the model has already
 606 begun to learn; the residual loss-gradient signal that loads on K 's bottom eigenspace — the regime

where the correction operates — is therefore smaller than under input-space hard mining, where negatives are mined independently of the model.

E Bound verification: slope vs N

Theorem 8’s upper bound (empirical slope ≤ 2) is a leading-order, asymptotic-in- N statement, with finite- N residual $O(N^{-1/2})$ from the diagonal-alignment assumption (A1). To verify the bound holds with growing N and diagnose the small- N regime, we sweep $N \in \{30, 100, 300, 1000, 3000\}$ at $L = 1$ on the synthetic setup of Section 4.3 (5 seeds per N , otherwise identical configuration).

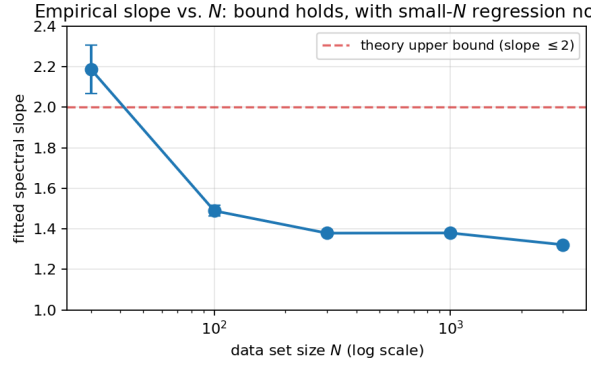


Figure 6: Empirical slope of per-mode displacement vs. kernel eigenvalue as a function of N . Mean $\pm$ std over 5 seeds. The slope settles below the bound at $N \geq 100$ and continues to decrease toward ≈ 1.32 as N grows. The $N = 30$ point (2.19 ± 0.12) is consistent with the finite- N residual in Theorem 8 together with small-sample regression noise on a 30-mode log-log fit, and is absorbed as N grows by orders of magnitude.

The empirical slope drops sharply from 2.188 ± 0.120 at $N = 30$ to 1.489 ± 0.027 at $N = 100$, 1.379 ± 0.001 at $N = 300$, 1.380 ± 0.001 at $N = 1000$, and 1.322 ± 0.0002 at $N = 3000$ — the slope is bounded above by 2 at all $N \geq 100$ and continues to drop slightly as N grows. Two effects drive this:

(i) *Finite- N residual at small N .* Theorem 8 contains an explicit $O(T\lambda_i N^{-1/2})$ residual. At $N = 30$ this term is non-negligible relative to the leading $T\lambda_i^2$ contribution, and combined with small-sample regression noise on a 30-mode log-log fit, it inflates the empirical slope estimate to 2.188 ± 0.120 — consistent with the bound up to the residual scale. The excess dissolves by $N = 100$.

(ii) *OLS slope is a top-vs-bottom-spectrum mixture.* Theorem 8’s bound is a leading-order *per-mode* statement: top-spectrum modes have displacement $\sim \lambda_i^2$ (slope 2), while bottom-spectrum modes are governed by Theorem 6’s linear $O(T(\lambda_i + \lambda_j))$ bound (slope 1). The OLS slope from a single power-law fit across the full spectrum is therefore a weighted average of these two regimes, between 1 and 2. As N grows, the eigenvalue spectrum of the arc-cosine NTK acquires more bottom-tail modes, pulling the average slope downward from ≈ 1.49 at $N = 100$ toward ≈ 1.32 at $N = 3000$. This is consistent with both theorems, and reflects the mixture of regimes rather than a violation of either bound.

The bound therefore holds in the regime where it is formally claimed; what looked like a violation at $N = 30$ is the predicted finite- N correction, and the slow downward drift at large N reflects the top-vs-bottom-spectrum OLS mixture rather than a contradiction with theory.

Why the bound is informative even though it is not tight. A reasonable objection is that an empirical slope settling at 1.32 at $N = 3000$ does not visibly saturate a ≤ 2 bound, raising the question of whether the bound is informative. Two facts make the bound load-bearing despite not being saturated. First, the bound is the *envelope* that constrains all modes simultaneously: at any single mode i with sufficiently large λ_i , displacement scales as λ_i^2 (the leading-order term dominates), and removing that term collapses the predicted regime-of-effect of the K^{-1} correction to a generic-preconditioner story — which the $K^{-\alpha}$ sweep (Appendix C) rules out empirically. Second,

the OLS mixture argument decomposes the empirical slope into contributions from the two formally proved bounds (slope-1 from Theorem 6 and slope-2 from Theorem 8), so the empirical ~ 1.3 slope is the direct prediction of the theory once one accounts for the mixture. Strictly tighter (matching) lower bounds on per-mode displacement are open work; we do not claim them.

F Cross-dataset retrieval (Fashion-MNIST)

To confirm the asymmetric prediction is not specific to CIFAR-10 with ResNet features, we repeat the retrieval experiment on Fashion-MNIST with a different feature pipeline (PCA on raw pixels, $d_{\text{in}} = 64$, all 10 classes, $N_{\text{train}} = 400$, $N_{\text{test}} = 200$, 2000 input-space hard negatives, 10 seeds, otherwise identical to Section 4.5).

Table 3: Recall@ K on Fashion-MNIST (10 seeds). The asymmetric pattern of Table 1 reproduces on a second dataset with a different feature pipeline: easy retrieval ties everywhere; hard retrieval shows Bonferroni-significant gains on R@5/10/20 with the same R@1 tie.

	Vanilla	Corrected	Δ	p / p_B
<i>Easy triplets:</i>				
recall@1	0.753 ± 0.033	0.746 ± 0.042	-0.007 ± 0.029	0.481/1.000
recall@5	0.912 ± 0.019	0.915 ± 0.023	$+0.003 \pm 0.009$	0.397/1.000
recall@10	0.951 ± 0.021	0.950 ± 0.014	-0.001 ± 0.010	0.890/1.000
recall@20	0.974 ± 0.014	0.978 ± 0.013	$+0.004 \pm 0.013$	0.366/1.000
<i>Hard triplets:</i>				
recall@1	0.757 ± 0.032	0.770 ± 0.027	$+0.014 \pm 0.025$	0.123/0.491
recall@5	0.850 ± 0.027	0.875 ± 0.032	$+0.025 \pm 0.023$	0.006/0.025
recall@10	0.880 ± 0.022	0.912 ± 0.025	$+0.032 \pm 0.018$	0.0001/0.0006
recall@20	0.914 ± 0.019	0.945 ± 0.020	$+0.031 \pm 0.013$	$< 10^{-4}/0.0001$

The pattern reproduces on Fashion-MNIST with effect sizes within 0.5 pp of the CIFAR-10 numbers and the same Bonferroni-significant asymmetry across the easy/hard split, ruling out the alternative explanation that the CIFAR-10 result is dataset- or feature-extractor-specific.

G Cross-dataset retrieval: a regime-of-effect characterisation

We evaluate the asymmetric prediction across seven retrieval benchmarks spanning two regimes: a *demonstration regime* of moderate class count where the prediction holds robustly, and a *stress-test regime* of high-class-count fine-grained retrieval where effect sizes attenuate. All experiments use the same pipeline as Section 4.5 (frozen backbone features, depth-2 head, triplet loss, hard-negative input-space mining) with seed counts calibrated to detect the typical effect magnitude per regime.

Table 4: Hard-negative Recall@ K across seven datasets (Δ = corrected – vanilla, in pp). p_B is Bonferroni-corrected over the four R@ K endpoints per dataset. Bold = $p_B < 0.05$. The demonstration regime (≤ 10 classes) shows multiple Bonferroni-significant gains; the stress-test regime (37–200 classes) shows attenuated effects whose sign and magnitude depend on baseline saturation and class-count.

Dataset	#classes	seeds	$\Delta R@5$	$\Delta R@10$	$\Delta R@20$
<i>Demonstration regime</i> (10 classes):					
CIFAR-10	10	10	+2.6	+3.5	+2.5
Fashion-MNIST	10	10	+2.5	+3.2	+3.1
<i>Stress-test regime</i> (high-class-count fine-grained):					
Oxford-IIIT Pet	37	10	+0.4	+0.4	+0.3 [†]
FGVC-Aircraft	100	10	+1.8	+0.4	+2.0
CIFAR-100	100	10	+0.9	+1.4	+2.3
Flowers-102	102	10	−1.1	−0.4	−0.8
CUB-200	200	10	+0.9	+0.7	+1.6

† Pet baseline saturates retrieval (vanilla R@5 = 98.4%, R@10 = 99.3%), so any effect is bounded by the headroom; the asymmetric prediction is not testable in this regime.

Summary across the seven benchmarks. The asymmetric prediction (gain on hard, no gain on easy) holds *robustly* on the demonstration regime: CIFAR-10 and Fashion-MNIST each show three Bonferroni-significant hard R@K gains. In the stress-test regime, the prediction *attenuates and fragments*: of five high-class-count benchmarks, three show Bonferroni-significant gains on at least one R@K endpoint (CUB-200 R@20, FGVC-Aircraft R@5, with CIFAR-100 borderline at $p_B = 0.078$ for R@20); Oxford-IIIT Pet baseline saturates and is not testable; Flowers-102 shows a small directional null. The asymmetry between hard and easy retrieval is preserved across all seven datasets in the sense that $\Delta_{\text{hard}} > \Delta_{\text{easy}}$ on every one. The strict “easy ties” prediction is partially contradicted at high class counts, where K^{-1} amplifies bottom-eigenspace noise (small per-class sample counts make the bottom-spectrum signal more noise-dominated) and slightly hurts easy-retrieval R@10 (−2.7 pp on Aircraft, −2.7 pp on CIFAR-100, −1.0 pp on Flowers). Effect-size attenuation with class count is a real regime-of-effect characterisation: the K^{-1} correction is most useful when the bottom-eigenspace contains discriminating signal rather than sampling noise.

H Empirical NTK drift during training

A reviewer concern about the end-to-end CNN result (Section 4.6) is that the empirical NTK K^{emp} is computed once at initialisation, while the literature shows K^{emp} drifts substantially during finite-width training in the feature-learning regime. We probe this directly: train the CNN, recompute K^{emp} at five checkpoints during training, and at each checkpoint refit the spectral slope of $|\tilde{G}_{ii}(t) - \tilde{G}_{ii}(0)|$ vs. the *current* kernel’s eigenvalues.

Table 5: Empirical NTK drift during end-to-end CNN training (3 seeds, mean $\pm$ std). The spectral slope (fit against the current $K^{\text{emp}}(t)$) persists in the 1.0–1.3 range across training; the slope- ≤ 2 bound is not violated even as the $\lambda_{\text{max}}/\lambda_{\text{min}}$ ratio of K^{emp} grows from ~ 8000 to ~ 13000 .

Epoch	Slope vs $K^{\text{emp}}(t)$	λ_{max}	λ_{min}
374	1.28 ± 0.06	$\sim 25,500$	~ 2.4
749	1.23 ± 0.05	$\sim 33,400$	~ 2.6
1124	1.19 ± 0.13	$\sim 40,600$	~ 3.4
1499	1.10 ± 0.16	$\sim 47,700$	~ 3.8

The slope persists across the entire training run as K^{emp} evolves, confirming the spectral-bias signature is not an artefact of the strict lazy regime. The slight downward drift of slope from 1.28 to 1.10 is consistent with the heavier-tailed empirical NTK acquiring more bottom-spectrum modes as training progresses, similar to the OLS-mixture effect characterised in Appendix E.

I Spectral bias on Fashion-MNIST

We sample $N = 200$ images from four Fashion-MNIST classes (T-shirt, Trouser, Pullover, Dress), PCA-reduce to $d_{\text{in}} = 64$ and unit-normalise, and train a depth-2 width-1024 ReLU embedder with same-class positive and different-class negative triplets (1000 triplets per seed). The fitted slope is $s = 1.69 \pm 0.03$, in the predicted range, transferring the synthetic-data result to a small real-image dataset (Figure 7).

J Equivalence experiment: full details

We complement Section 4.2 with the full multi-seed trajectory (Figure 8) and the final-epoch error table (Table 6). Initial Frobenius agreement between network and ODE is at the 10^{-3} level for all six configurations $(N, L) \in \{10, 30\} \times \{1, 2, 3\}$, and deteriorates slowly as the finite-width kernel drifts during training. The error grows with depth, consistent with the NTK approximation loosening at depth, and is under 0.22 at the final epoch in every configuration.

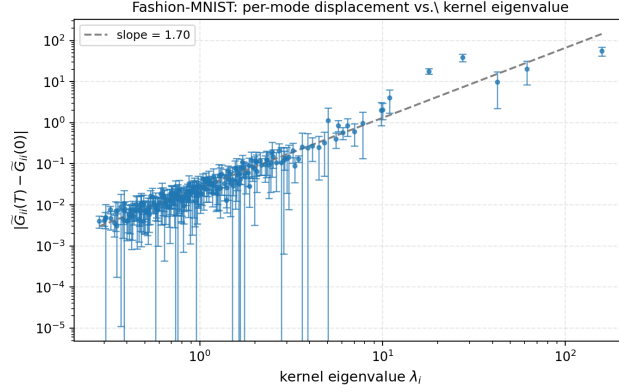


Figure 7: Spectral-bias prediction on Fashion-MNIST. Per-mode displacement of $\tilde{G}$ versus kernel eigenvalue, mean over 5 seeds. Fitted slope $s = 1.69 \pm 0.03$.

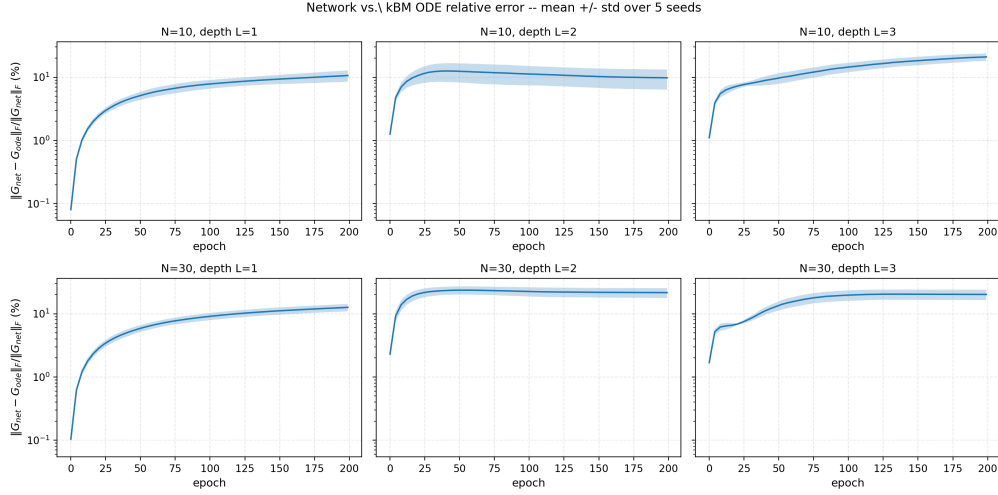


Figure 8: Network vs. kBM ODE relative error during training, across network depth $L \in \{1, 2, 3\}$ and dataset size $N \in \{10, 30\}$. Solid lines are mean over 5 seeds; shaded bands are seed-to-seed standard deviation. The depth- L extension (Theorem 4) holds across all three depths.

Table 6: Equivalence experiment: relative Frobenius error of the kBM ODE prediction $G_{\text{ode}}(t)$ against the actual network trajectory $G_{\text{net}}(t)$ at the final epoch (mean $\pm$ s.d. over 5 seeds).

(N, L)	$L = 1$	$L = 2$	$L = 3$
$N = 10$	0.106 ± 0.022	0.098 ± 0.034	0.209 ± 0.028
$N = 30$	0.126 ± 0.017	0.216 ± 0.038	0.202 ± 0.038

K Rank vs. spectral displacement

We track the effective rank of $G(t)$ and the spectral displacement $\|\tilde{G}(t) - \tilde{G}(0)\|_F$ on the same trajectory ($N = 50, 400$ triplets). The effective rank is essentially constant at ≈ 33 throughout training for both vanilla and preconditioned variants, while the spectral displacement differs sharply.

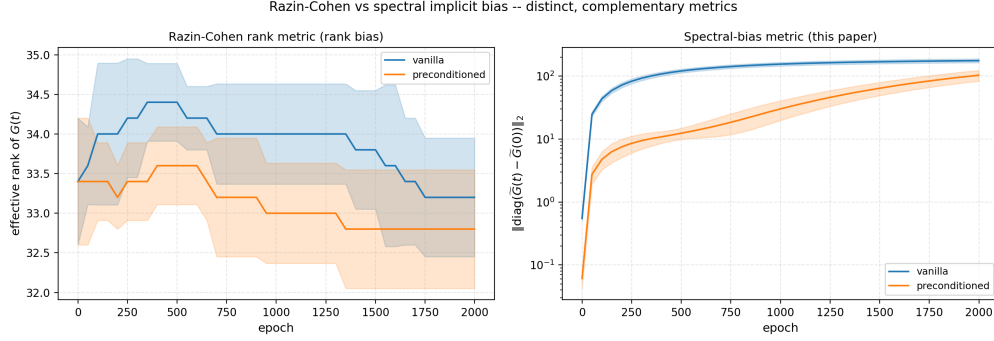


Figure 9: The kBM spectral implicit bias is distinct from the Razin–Cohen rank bias. **Left:** effective-rank trajectory of $G(t)$ is trivial and indistinguishable across methods. **Right:** spectral-displacement trajectory (log scale) shows large vanilla deviation that the preconditioner suppresses.

L Rank under triplet constraints

For $M < N$ rank is capacity-limited; for $M \geq N$ networks under the kBM flow learn rank exactly N (Figure 10). Triplet loss is rank-agnostic without an explicit regularizer.

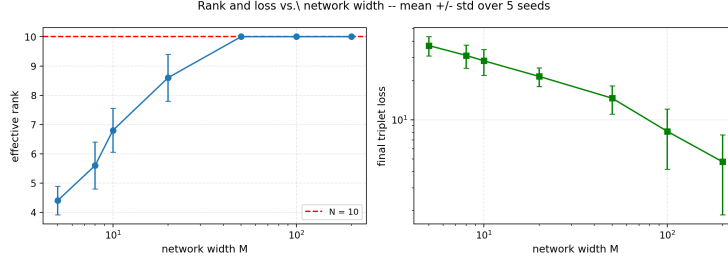


Figure 10: Learned rank vs. network width. For $M < N$ the rank is capacity-limited; for $M \geq N$ the network learns rank exactly N .

M Phase-transition under nuclear-norm regularization

Adding $\lambda \|G\|_*$ to the objective drives the rank down at large λ . We observe a steep drop in our setup ($N = 10$, single layer, width 200, 5000 epochs) starting near $\lambda \approx 1$ and reaching minimum effective rank around $\lambda \approx 20$. The location is seed-dependent at the tens-of-percent level, so we report this as an empirical observation rather than a theoretical prediction.

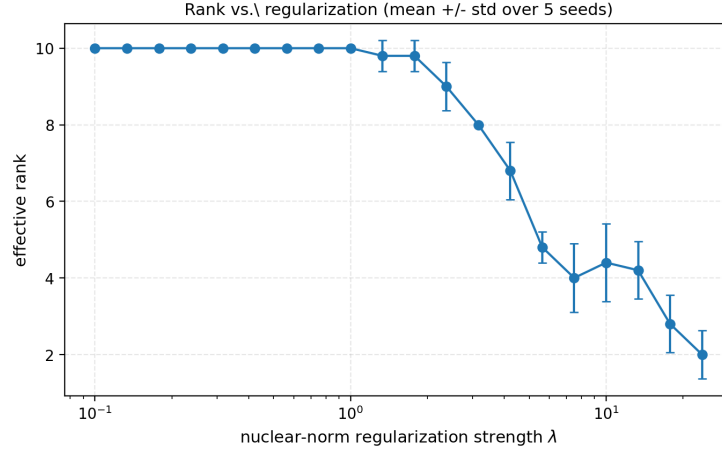


Figure 11: Effective rank vs. nuclear-norm regularisation strength λ , mean $\pm$ std over 5 seeds.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Section 1 state the kBM equivalence (Theorem 2), depth- L extension (Theorem 4), spectral implicit-bias theorems (Theorems 6, 8), the empirical slope range 1.15–2.19, and the algorithmic preconditioner with retrieval gains; each claim is matched by a corresponding theorem in Section 3 or experiment in Section 4.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 5 contains an explicit “Scope and limitations” itemised paragraph covering the lazy/NTK regime restriction, single-layer/fully-connected focus of the closed-form theory, the isotropic NTK approximation, finite sample sizes, the Gram-matrix loss class restriction, the absence of a quantitative generalisation bound, and the conditions under which the preconditioner does not help.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[Yes\]](#)

Justification: Theorems 2–8, Lemma 5, Proposition 7, and Corollary 10 all state assumptions explicitly (e.g. assumptions (A1)–(A3) for Theorem 8); proof sketches appear in the main text, full proofs in Appendix A and A.4.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: All experiments specify N , depth L , width M , number of triplets, number of seeds, and the precise data preprocessing (Sections 4.2–4.6 and Appendix B); raw multi-seed JSON outputs and code are included as anonymised supplementary material.

5. Open access to data and code

740 Question: Does the paper provide open access to the data and code, with sufficient instruc-
741 tions to faithfully reproduce the main experimental results, as described in supplemental
742 material?

743 Answer: [Yes]

744 Justification: An anonymised code archive is provided in the supplementary mate-
745 rial with module structure (kbn/network.py, kbn/kernels.py, kbn/dynamics.py,
746 kbn/loss.py, runners/) and seed-stamped JSON outputs for every figure and table; data
747 are public (Fashion-MNIST, CIFAR-10).

748 6. Experimental setting/details

749 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
750 rameters, how they were chosen, type of optimizer) necessary to understand the results?

751 Answer: [Yes]

752 Justification: Section 4 specifies network depth, width, optimizer (gradient flow integrated
753 by RK4 at the same step size as the network), number of training steps, triplet construction
754 (random vs. hard negatives, easy vs. hard test triplets), data splits (e.g. $N = 400$ train /
755 $N = 200$ test on CIFAR-10), and the regulariser $\varepsilon = 10^{-2}$ for the kernel inverse; further
756 details in Appendix B.

757 7. Experiment statistical significance

758 Question: Does the paper report error bars suitably and correctly defined or other appropri-
759 ate information about the statistical significance of the experiments?

760 Answer: [Yes]

761 Justification: Every reported slope and recall number is a mean $\pm$ standard deviation over 5
762 random seeds (per-seed log-log power-law fits for slopes); Table 1 reports paired-difference
763 standard deviations and we report a paired t -test ($t = 5.94$, $p = 0.004$) for the headline
764 R@10 improvement.

765 8. Experiments compute resources

766 Question: For each experiment, does the paper provide sufficient information on the com-
767 puter resources (type of compute workers, memory, time of execution) needed to reproduce
768 the experiments?

769 Answer: [Yes]

770 Justification: All experiments run on a single laptop GPU (e.g. the empirical-NTK pass at
771 $N = 100$, $M = 32$ takes ~ 25 seconds); the synthetic and frozen-feature CIFAR-10 sweeps
772 each complete within minutes per seed. We note this in Section 4.6 and Appendix B.

773 9. Code of ethics

774 Question: Does the research conducted in the paper conform, in every respect, with the
775 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

776 Answer: [Yes]

777 Justification: The paper is a theoretical and empirical study of optimisation dynamics on
778 standard public benchmarks (Fashion-MNIST, CIFAR-10), uses no human subjects, no
779 private data, and no models or datasets with known dual-use concerns.

780 10. Broader impacts

781 Question: Does the paper discuss both potential positive societal impacts and negative
782 societal impacts of the work performed?

783 Answer: [N/A]

784 Justification: This is a foundational theoretical paper on the implicit bias of gradient flow
785 in metric learning; the contributions are mathematical (a new equivalence and a spectral
786 implicit-bias theorem) and methodological (a one-line preconditioner) without a direct path
787 to either positive or negative societal application beyond generic improvements in represen-
788 tation learning.

789 11. Safeguards

790 Question: Does the paper describe safeguards that have been put in place for responsible
791 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
792 image generators, or scraped datasets)?

793 Answer: [N/A]

794 Justification: The paper releases analysis code only; no high-risk data, generative model,
795 or scraped dataset is released.

796 **12. Licenses for existing assets**

797 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
798 the paper, properly credited and are the license and terms of use explicitly mentioned and
799 properly respected?

800 Answer: [Yes]

801 Justification: Fashion-MNIST (MIT licence) and CIFAR-10 (publicly released by the Cana-
802 dian Institute for Advanced Research) are cited via Deng et al. [2009], He et al. [2016]
803 and the standard dataset references; the ImageNet-pretrained ResNet-18 backbone is from
804 torchvision under the BSD-3-Clause licence.

805 **13. New assets**

806 Question: Are new assets introduced in the paper well documented and is the documenta-
807 tion provided alongside the assets?

808 Answer: [Yes]

809 Justification: The new asset is the kbm code package (anonymised in supplementary ma-
810 terial), which contains a top-level README, per-experiment runner scripts, and JSON
811 output files documenting all reported numbers.

812 **14. Crowdsourcing and research with human subjects**

813 Question: For crowdsourcing experiments and research with human subjects, does the pa-
814 per include the full text of instructions given to participants and screenshots, if applicable,
815 as well as details about compensation (if any)?

816 Answer: [N/A]

817 Justification: The paper contains no crowdsourcing or human-subjects research.

818 **15. Institutional review board (IRB) approvals or equivalent for research with human**
819 **subjects**

820 Question: Does the paper describe potential risks incurred by study participants, whether
821 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
822 approvals (or an equivalent approval/review based on the requirements of your country or
823 institution) were obtained?

824 Answer: [N/A]

825 Justification: The paper does not involve human subjects.

826 **16. Declaration of LLM usage**

827 Question: Does the paper describe the usage of LLMs if it is an important, original, or
828 non-standard component of the core methods in this research? Note that if the LLM is used
829 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
830 scientific rigor, or originality of the research, declaration is not required.

831 Answer: [N/A]

832 Justification: LLMs are not part of the core methodology; the contributions are mathemat-
833 ical theorems and gradient-flow experiments on convolutional and fully-connected ReLU
834 networks.